

Intro to DSP - MUSIC 320 //Homework 2 (100 Points)

Handed out on: October 6th, 2025 - 3:30 pm

Due on: October 20th, 2025 - 1:30 pm

Submission instructions

The homework must be submitted using Canvas's assignment submission system (in the Assignments tab). Select the associated link in that tab and proceed with your submission.

You must **turn in 2 items**:

- **A write-up (with all the instructed answers and figures)** in PDF format. Scanned and typed submission are both acceptable.
- **All your code and data files in a single compressed file** (zip, tar or tar.gz) containing the jupyter notebook with all the answers to the lab exercises and any other files requested in the homework instructions.

For your files, use the following **naming convention**: hwX.suid.type where X is the homework number and suid is your Stanford ID. For example, the write-up for homework 1 will be hw1.mab.pdf and the code files hw1.mab.zip/hw1.mab.tar.gz/hw1.mab.tar.

Theory Problems (65 pts.)

1. (10 pts.) Inner Product:

- (a) Consider the vectors $\vec{v} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Please draw them in the x-y plane and determine the angle between them. Remember the definition of the inner product as matrix multiplication: $\langle \vec{v} | \vec{w} \rangle = (\vec{v})^\dagger \vec{w}$ where the dagger complex conjugates and turns a column vector into a row vector.
- (b) Draw the projection of $\vec{w}$ onto $\vec{v}$ and vice-versa. What are the lengths of each of these vectors and what are the lengths of their projections onto each other?
- (c) What are the inner products $\langle \vec{v} | \vec{w} \rangle$ and $\langle \vec{w} | \vec{v} \rangle$? How are the two inner products related?
- (d) Now consider the vectors $\vec{z}_1 = \begin{bmatrix} 0 \\ j \\ 1 \end{bmatrix}$ and $\vec{z}_2 = \begin{bmatrix} 1 \\ j \\ j \end{bmatrix}$. What are the inner products $\langle \vec{z}_1 | \vec{z}_2 \rangle$ and $\langle \vec{z}_2 | \vec{z}_1 \rangle$? How are the two inner products related?

2. (10 pts.) Gram-Schmidt Orthogonalization:

Consider the three vectors $\vec{w}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\vec{w}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $\vec{w}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

- (a) Are they independent? (Can you show that there are no real c_1, c_2, c_3 that aren't all zero which make $c_1\vec{w}_1 + c_2\vec{w}_2 + c_3\vec{w}_3 = 0$? Hint: Start at the 3rd equation.)

- (b) Are the three $\vec{w}_k$ orthogonal? (Recall that two vectors are orthogonal if their inner product is zero.) If not, please use Gram-Schmidt to make them orthogonal.

3. (10 pts.) Orthonormal Bases:

Suppose we have a real 3-vector $\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ v_2 \end{bmatrix} = \hat{e}_0 v_0 + \hat{e}_1 v_1 + \hat{e}_2 v_2$ where $\hat{e}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{e}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, $\hat{e}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

Consider the three vectors $\vec{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $\vec{w}_2 = \begin{bmatrix} 2 \\ -1 \\ -1 \end{bmatrix}$, $\vec{w}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$.

- Are the three $\vec{w}_k$ independent? (Can you show that there are no real c_1, c_2, c_3 that aren't all zero which make $c_1 \vec{w}_1 + c_2 \vec{w}_2 + c_3 \vec{w}_3 = 0$?)
- Are the three $\vec{w}_k$ orthogonal? If not, please use Gram-Schmidt to make them orthogonal.
- Please compute the norms of each of the orthogonal vectors from step (b) and convert them into an orthonormal basis $\hat{b}_k$.
- Please create the coordinate transformation matrix (from the $\hat{e}_i$ to the $\hat{b}_k$) and apply it to the vector $\vec{v}$ to get $\vec{V}$ written in the $\hat{b}_k$ coordinates. (Recall that the k^{th} row of this matrix is the complex conjugate of $\hat{b}_k$ written in the $\hat{e}_i$ basis.)

4. (20 pts.) Orthonormal Bases:

Suppose we have a real 4-vector $\vec{v} = \begin{bmatrix} v_0 \\ v_1 \\ v_2 \\ v_3 \end{bmatrix} = \hat{e}_0 v_0 + \hat{e}_1 v_1 + \hat{e}_2 v_2 + \hat{e}_3 v_3$

where $\hat{e}_0 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, $\hat{e}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, $\hat{e}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$, $\hat{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.

Consider the four vectors $\vec{w}_k$ for $k = 0, \dots, 3$ where the n^{th} element of $\vec{w}_k$ is equal to $\exp^{j2\pi(\frac{kn}{4})}$ for $n = 0, \dots, 3$.

- Please compute the sum $\sum_{k=0}^3 \vec{w}_k$.
- Are the four $\vec{w}_k$ independent? (Can you show that there are no complex c_0, c_1, c_2, c_3 that make $c_0 \vec{w}_0 + c_1 \vec{w}_1 + c_2 \vec{w}_2 + c_3 \vec{w}_3 = 0$ without them all being zero? Hint: Try adding the 1st and 3rd equations and the 2nd and 4th equations in the case where you assume $c_0 \neq 0$. In the case where $c_0 = 0$, start by adding the 1st and 3rd equations to see what that tells you.)
- Are the four $\vec{w}_k$ orthogonal? If not, please use Gram-Schmidt to make them orthogonal.
- Please compute the norms of each of the orthogonal vectors from step (c) and convert them into an orthonormal basis $\hat{b}_k$.
- Please create the coordinate transformation matrix (from the $\hat{e}_i$ to the $\hat{b}_k$) and apply it to the vector $\vec{v} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$ to get $\vec{v}$ written in the coordinates $\hat{b}_k$ (which you can denote $\vec{V}$). (Recall that the k^{th} row of this matrix is the complex conjugate of $\hat{b}_k$ written in the $\hat{e}_i$ basis.)
- Please use the coordinate transformation matrix from (e) to write the basis vectors $\hat{e}_i$ in the basis $\hat{b}_k$. (Hint: Think about how the answer relates to the columns of the transformation matrix and this will be easy!)
- Please create the coordinate transformation matrix from the $\hat{b}_k$ to the $\hat{e}_i$ and apply it to the vector $\vec{V}$. Do you recover $\vec{v}$? (Recall that the i^{th} row of this matrix is the complex conjugate of $\hat{e}_i$ written in the $\hat{b}_k$ basis and keep in mind that you just wrote $\hat{e}_i$ in the basis $\hat{b}_k$ for part (f).)

5. (15 pts.) Compute the N=8 DFT

(a) Write down the forward transform matrix for the N=8 DFT.

(b) Compute the N=8 DFT of the following 8-sample signals:

i. $x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

ii. $x = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

iii. $x = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Lab Assignments (35 Points) Please see the attached Jupyter notebook.